PHYSIOLOGY · MCQ PRACTICE

GUE BLOCK

Acid–Base Balance by the Kidneys — MCQ Practice

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Physiology GUE Block · MEDucation by Mattin Majid

HOW TO USE THIS SET

20 questions, 5 options each (A–E). Answer on paper or in your head, then check every answer against the key on the final page.

4 Easy · 10 Medium · 6 Hard · suggested time ≈ 25 minutes.

Covers pH/Henderson-Hasselbalch, renal bicarbonate reclamation vs new bicarbonate generation, urinary buffers, net acid excretion, and compensation rules.

Q Questions

1. What is the normal range for arterial pH? EASY

A. 7.00–7.20
B. 7.15–7.30
C. 7.35–7.45
D. 7.45–7.60
E. 7.60–7.80

2. Which organ removes CO_2 from the body, and by what process? EASY

A. Kidneys, by filtration
B. Liver, by metabolism
C. Lungs, by exhalation
D. Skin, by sweating
E. Intestine, by absorption

3. A molecule that can accept a hydrogen ion (H^+) is called a: EASY

A. Acid
B. Base
C. Buffer salt only
D. Oxidant
E. Electrolyte

4. A fall of one pH unit corresponds to what change in $[H^+]$? EASY

A. No change
B. A twofold rise
C. A tenfold rise
D. A hundredfold rise
E. A tenfold fall

5. Using the Henderson-Hasselbalch equation, what is the approximate plasma pH when [HCO3-] is 24 mmol/L and PCO2 is 40 mmHg? MEDIUM

A. 6.10
B. 7.00
C. 7.40
D. 7.80
E. 8.00

6. Which defense against pH change acts fastest? MEDIUM

A. The kidneys, within seconds
B. The lungs, within seconds
C. Chemical buffers, within seconds
D. The kidneys, within minutes
E. The lungs, within hours to days

7. Where does the proximal tubule's HCO3- reclamation mechanism begin, at the apical membrane? MEDIUM

A. AE1 chloride-bicarbonate exchange
B. NHE3-mediated H+ secretion into the lumen
C. Pendrin-mediated bicarbonate secretion
D. H+/K+-ATPase on the apical membrane
E. Aquaporin-2 insertion

8. What is the approximate fraction of filtered bicarbonate reclaimed by the proximal tubule? MEDIUM

A. 10–15%
B. 25–30%
C. 50%
D. 80–90%
E. 100% only in alkalosis

9. Which statement correctly describes 'new bicarbonate' generation, as distinct from reclamation? MEDIUM

A. It occurs when filtered HCO3- combines with secreted H+ and returns to blood unchanged
B. It occurs when secreted H+ is buffered by a nonbicarbonate buffer (phosphate or ammonia) in the urine, adding a new HCO3- equivalent to the blood
C. It only occurs in the proximal tubule
D. It never involves ammonium
E. It decreases net acid excretion

10. Which basolateral transporter in the alpha-intercalated cell exchanges bicarbonate for chloride, sending HCO3- to the blood? MEDIUM

A. NHE3
B. NBCe1
C. AE1
D. Pendrin
E. NKCC2

11. During metabolic alkalosis, which cell type and apical transporter secretes excess bicarbonate into the urine? MEDIUM

A. Alpha-intercalated cell, via H+-ATPase
B. Beta-intercalated cell, via apical pendrin
C. Proximal tubule cell, via NHE3
D. Principal cell, via aquaporin-2
E. Thick ascending limb cell, via NKCC2

12. Why can the kidneys excrete only a limited amount of acid as titratable acid (phosphate) alone? MEDIUM

A. Phosphate buffering capacity is limited by the amount of filtered phosphate available
B. Titratable acid excretion requires no urinary buffer at all
C. Phosphate is not filtered at the glomerulus
D. Titratable acid excretion is unlimited and always sufficient
E. Phosphate only buffers in the blood, not urine

13. Which metabolic pathway increases substantially during acidosis to boost ammonium production in the proximal tubule? MEDIUM

A. Beta-oxidation of fatty acids
B. Glutamine metabolism
C. Glycogenolysis
D. Urea cycle in the kidney
E. Lactate dehydrogenase pathway

14. A patient's labs show Na+ 140, Cl- 116, HCO3- 14 mmol/L. What is the anion gap, and is it elevated? MEDIUM

A. 4 mmol/L — low
B. 10 mmol/L — within the usual range
C. 26 mmol/L — elevated
D. 130 mmol/L — elevated
E. 154 mmol/L — elevated

15. Which factor can suppress renal ammoniagenesis and impair acid excretion, as seen in type 4 renal tubular acidosis? HARD

A. Low plasma potassium
B. High plasma potassium
C. Low plasma sodium
D. High plasma bicarbonate only
E. Low plasma chloride

16. A patient has diarrhea with pH 7.29, HCO3- 14 mmol/L, PCO2 30 mmHg, Na+ 140, Cl- 116. Using Winter's rule (PCO2 = 1.5xHCO3- + 8 +/- 2), is the respiratory response appropriate, and what is the anion gap? HARD

A. Expected PCO2 is 29+/-2 (27-31); observed 30 fits; anion gap = 10, within range — appropriate compensation with a non-gap picture
B. Expected PCO2 is 40; observed 30 indicates an additional respiratory alkalosis
C. Anion gap is 26, indicating a gap acidosis superimposed
D. The respiratory response is absent entirely
E. Winter's rule does not apply to diarrhea-induced acidosis

17. A patient has vomiting-induced metabolic alkalosis with low effective circulating volume and low plasma chloride and potassium. Despite elevated plasma HCO3-, the kidneys fail to excrete the excess bicarbonate as expected. Which mechanism best explains this persistence? HARD

A. Low volume increases angiotensin II, which increases proximal Na+ and HCO3- reclamation, maintaining the alkalosis despite elevated HCO3-
B. Low volume has no effect on proximal HCO3- handling
C. High chloride is driving continued HCO3- retention
D. Aldosterone is suppressed in this state, preventing HCO3- excretion
E. The thick ascending limb directly excretes excess HCO3- regardless of volume status

18. A patient has chronic respiratory acidosis (long-standing hypoventilation) with pH 7.35, PCO2 60 mmHg, HCO3- 32 mmol/L. Compare this to a patient with acute respiratory alkalosis (brief hyperventilation): pH 7.49, PCO2 30 mmHg, HCO3- 22 mmol/L. What best explains why the chronic acidosis patient's HCO3- has risen so much more than would be expected acutely? HARD

A. Chronic respiratory acidosis allows time for renal adaptation — increased net acid excretion over days raises HCO3- substantially (≈3-4 mmol/L per 10 mmHg PCO2 rise), whereas the acute alkalosis reflects mainly early chemical buffering with no time for renal adaptation
B. Both patients show purely acute buffering with no renal contribution
C. The acidosis patient's kidneys are excreting more bicarbonate, not retaining it
D. Respiratory alkalosis always produces a larger HCO3- change than respiratory acidosis
E. Chronic acidosis prevents any renal response at all

19. A patient has a selective defect in collecting-duct Rhcg function (impairing NH3 transfer across the collecting duct) while proximal glutamine metabolism and NH4+ production remain entirely normal. What is the most likely consequence for net acid excretion? HARD

A. No consequence, since proximal NH4+ production is all that matters for net acid excretion

B. Impaired ammonium excretion despite normal production, since excretion — not production alone — determines the net systemic bicarbonate gain from the ammonium pathway

C. Increased titratable acid excretion fully compensates

D. Proximal HCO3- reclamation is directly impaired by this defect

E. Respiratory compensation eliminates any acid-base consequence

20. A patient's labs: pH 7.42, HCO3- 30 mmol/L, PCO2 46 mmHg. The pH is within the normal range. Why might this still represent a clinically important acid-base disturbance rather than normal physiology? HARD

A. A normal pH always rules out any acid-base disturbance

B. Both HCO3- and PCO2 are elevated above their usual baselines (24/40), suggesting two opposing primary processes (e.g. a metabolic alkalosis and a respiratory acidosis) are coexisting and partially offsetting each other's effect on pH

C. This pattern is only possible in healthy individuals

D. Elevated PCO2 alone always produces acidemia regardless of HCO3-

E. The anion gap must be calculated before any interpretation is possible, and no other inference can be drawn

✓ Answer key

Q	Answer	Difficulty	Why
1	C	EASY	Normal arterial pH is 7.35–7.45; acidemia is <7.35 and alkalemia is >7.45.
2	C	EASY	CO2 is removed by the lungs during exhalation, while the kidneys handle bicarbonate and H+ (nonvolatile acid).
3	B	EASY	A base is an ion or molecule that can accept H+, such as HCO3- or plasma proteins.
4	C	EASY	pH is a negative log10 scale, so each one-unit fall in pH represents a tenfold rise in hydrogen ion concentration.
5	C	MEDIUM	pH = 6.1 + log10(24/(0.03x40)) = 6.1 + log10(20) ≈ 6.1 + 1.3 = 7.40.
6	C	MEDIUM	Chemical buffers act within seconds; the lungs respond within minutes; the kidneys take hours to days.
7	B	MEDIUM	NHE3 secretes H+ into the lumen at the apical membrane, which combines with filtered HCO3- to begin the reclamation sequence.
8	D	MEDIUM	The proximal tubule reclaims roughly 80-90% of the filtered HCO3- load; the thick ascending limb handles most of the remainder (~10-15%).
9	B	MEDIUM	New bicarbonate generation happens when H+ is buffered by phosphate or ammonia (not filtered HCO3-), so the HCO3- generated intracellularly is a net addition to blood, unlike reclamation.
10	C	MEDIUM	AE1 (anion exchanger 1) is the basolateral Cl-/HCO3- exchanger in the alpha-intercalated cell that sends generated HCO3- to the blood.
11	B	MEDIUM	In alkalosis, beta-intercalated cells reverse polarity, using apical pendrin to secrete HCO3- (for Cl-) into the lumen.
12	A	MEDIUM	Phosphate is the major titratable buffer, but its urinary supply is limited by the filtered amount, capping how much acid can be excreted this way — hence the importance of the adaptive ammonium pathway.
13	B	MEDIUM	Acidosis increases proximal tubular glutamine metabolism, producing NH4+ (toward urine) and new HCO3- (toward blood).
14	B	MEDIUM	Anion gap = Na+ - (Cl- + HCO3-) = 140 - (116+14) = 10 mmol/L, within the usual laboratory range.
15	B	HARD	High plasma potassium suppresses ammoniagenesis, a mechanism implicated in type 4 renal tubular acidosis.
16	A	HARD	Winter's rule gives 1.5x14+8 = 29+/-2 (27-31 mmHg); observed PCO2 of 30 fits, confirming appropriate respiratory compensation. Anion gap = 140-(116+14) = 10, a usual non-gap (hyperchloremic) picture consistent with GI bicarbonate loss.

17	**A**	HARD	Low effective circulating volume activates angiotensin II, which increases proximal Na+/HCO3- reclamation; this volume-driven mechanism can maintain an alkalosis even though elevated HCO3- would otherwise be expected to be excreted. Chloride and potassium depletion further limit the kidney's ability to correct it.
18	**A**	HARD	Chronic respiratory acidosis gives the kidneys days to adapt, raising net acid excretion and HCO3- substantially (~3-4 mmol/L per 10 mmHg PCO2, versus ~1 acutely). Acute respiratory alkalosis, happening over minutes, reflects mainly immediate chemical buffering (~2 mmol/L fall per 10 mmHg PCO2 fall) without time for the renal adaptation that drives the larger chronic change.
19	**B**	HARD	The lecture specifies that for ammonium, urinary excretion must follow renal production for a net systemic bicarbonate gain. A defect in collecting-duct NH3 transfer (Rhcg) blocks final excretion even with normal proximal production, impairing the kidney's adaptive acid-excretion capacity despite normal ammoniagenesis.
20	**B**	HARD	A normal pH can conceal two opposing disturbances. Here both HCO3- and PCO2 are above their normal baseline values (24 and 40 respectively), which is not what simple compensation for a single primary process would produce — it suggests a mixed disorder (e.g., concurrent metabolic alkalosis and respiratory acidosis) whose opposing pH effects happen to leave pH near normal.